UNIVERSITY OF MANNHEIM
School of Social Sciences

UNIVERSITY OF MANNHEIM
EXPERIMENTAL PSYCHOLOGY LAB

An application of established computational models of quantitative estimation in three real-world domains

David Izydorczyk & Arndt Bröder

Theoretical Background

Quantitative estimation is a fundamental everyday cognitive ability, in which people assign numerical values to real-world objects (e.g., nutritional values, severity of symptoms etc.)

Different cognitive processes and mechanisms have been proposed to explain how people make such estimates, including rule-based integration of cues, similarity-based retrieval of exemplars, and heuristic strategies.[e.g., 1,2,3,4]

Research has largely followed two separate traditions: precise computational models tested with artificial stimuli[e.g., 1,2,3,4] versus studies using realistic tasks but less formal theoretical accounts.[e.g., 6, 7]

Aim: This study investigates whether **five established computational models** of estimation explain judgments in real-world domains and whether dominant cognitive processes vary across domains.

The Models

Rule-based/Cue Abstraction Model (CAM):[1] Assumes individuals estimate a criterion value by calculating a weighted linear sum of an object's individual features or cues according to an abstracted rule.

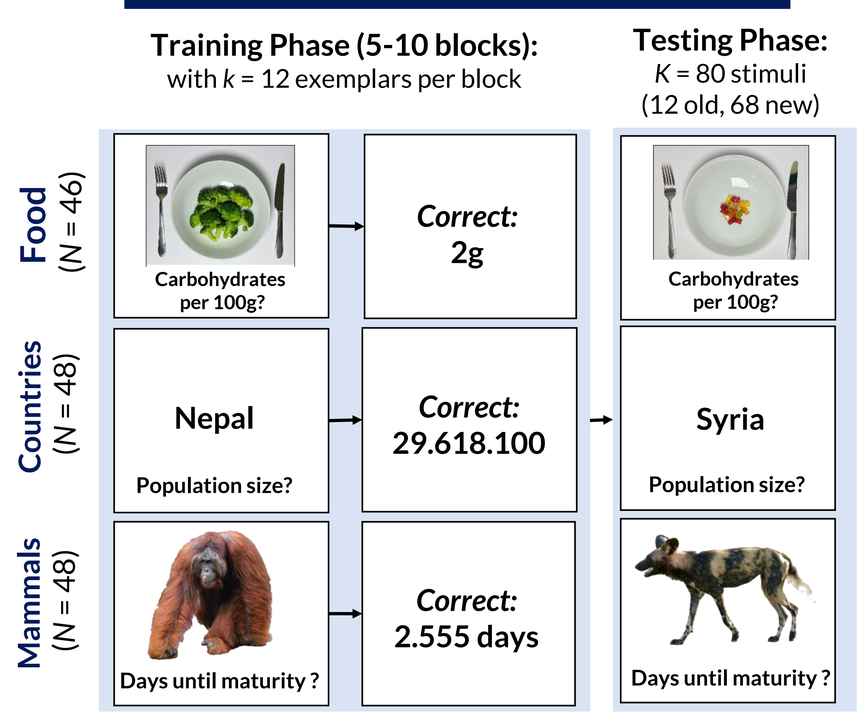
Exemplar-based/Generalized Context Model (GCM):[2,8] Assumes judgments are formed by taking a similarity-weighted average of criterion values from previously encountered objects stored in memory.

RulEx-J Model:[3] Assumes that estimation involves a continuous, parallel integration of both rule-based and exemplar-based processes

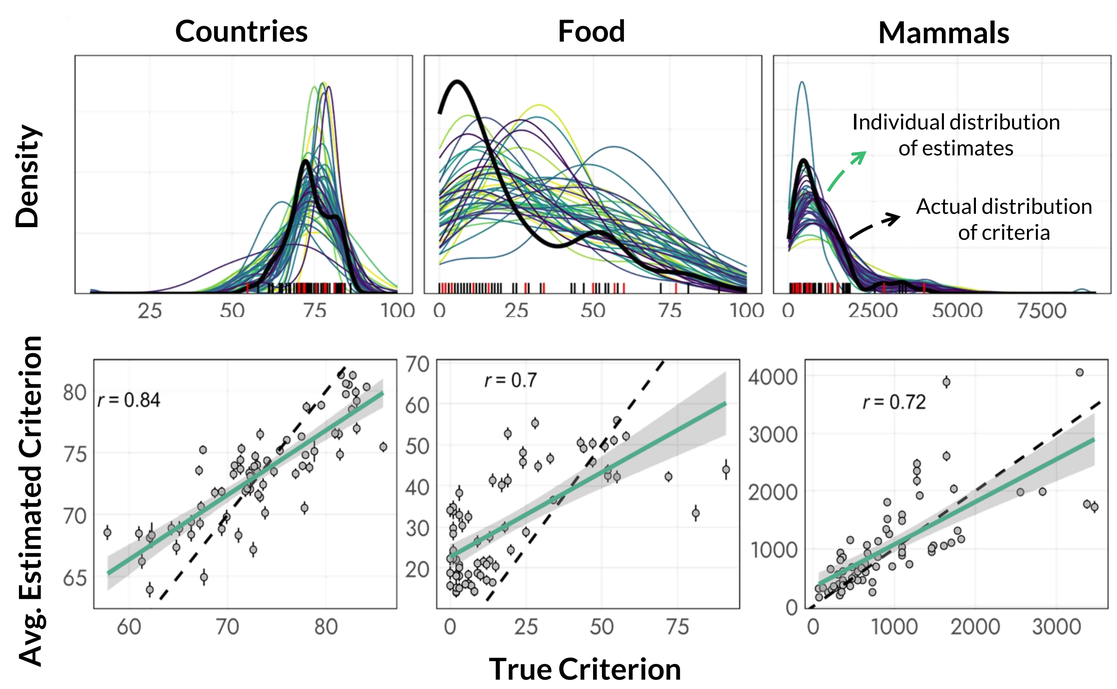
Mapping Model (MAPP): [4]Assumes people categorize objects based on their feature patterns and assign them the median criterion value of other items in that same category.

QuickEst Heuristic:[5] Assumes a "fast and frugal" strategy that processes cues sequentially and bases the estimate on the first cue for which an object lacks the corresponding property

Method



Behavioral Results



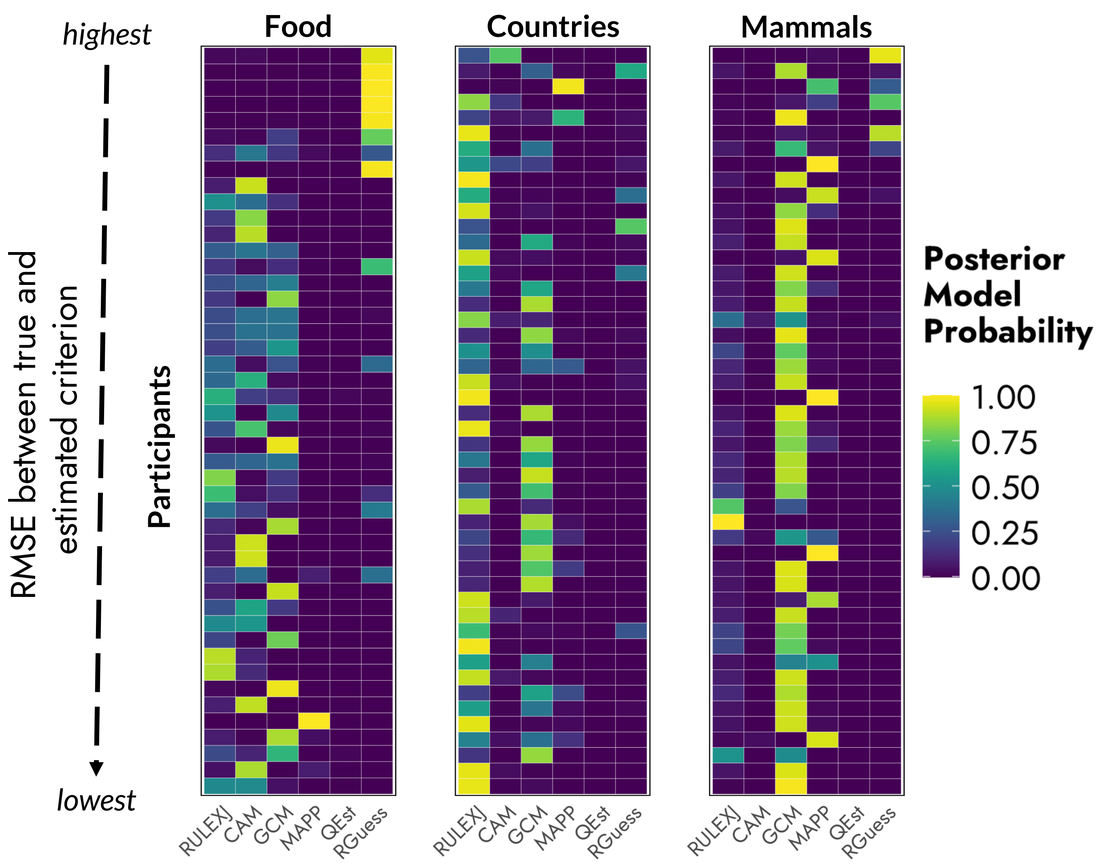
Summary:

- Participants successfully transferred knowledge from trained exemplars to novel objects across all domains.
- (Average) Estimates closely aligned with actual criteria distributions for mammals and countries.
- Overestimation in the food domain, likely due to the selection of exemplars

Modeling Results

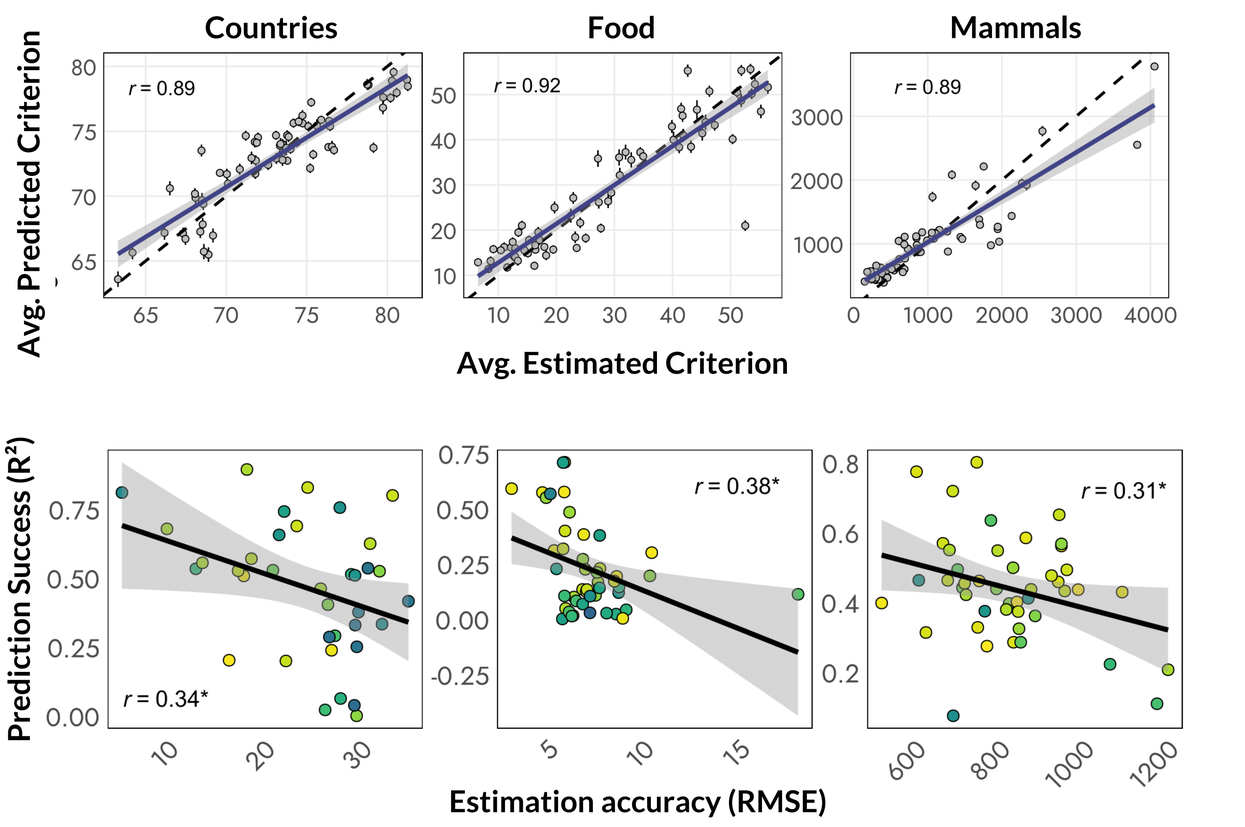
Relative model comparison

(via simulation-based inference with `Bayesflow`)



Absolute *average* and *individual* model fit

(based on median posterior predictions for each trial of the model with the highest posterior model probability)



Discussion

Ecological Validity of Computational Models: Results demonstrate that cognitive process models originally developed in controlled laboratory settings can successfully account for human estimation in complex, real-world domains.

Domain-Specific Process Dominance: No single cognitive process dominated across all domains; instead, process dominance varied systematically with domain structure, such as taxonomic organization or the linearity of cue-criterion relationships.

Structural Determinants of Judgment: Aligns with an ecological rationality perspective and past research, suggesting that individuals adapt their estimation strategies to match the environmental structure of the task.

Literature:

[1] Juslin, P., Olsson, H., & Olsson, A. C. (2003). Exemplar effects in categorization and multiple-cue judgment. *Journal of Experimental Psychology*, *132* (1), 133–156.
[2] Hoffmann, J. A., von Helversen, B., & Rieskamp, J. (2016). Similar task features shape judgment and categorization processes. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *42* (8), 1193–1217.
[3] Bröder, A., Gräf, M., & Kieslich, P. J. (2017). Measuring the relative contributions of rule-based and exemplar-based processes in judgment: Validation of a simple model. *Judgment and Decision Making*, *12* (5), 491–506.
[4] von Helversen, B., & Rieskamp, J. (2008). The mapping model: A cognitive theory of quantitative estimation. *Journal of Experimental Psychology*, *137* (1), 73–96.
[5] Hertwig, R., Hoffrage, U., & Martignon, L. (1999). Quick estimation: Letting the environment do the work. In G. Gigerenzer, P. M. Todd, & the ABC Research Group (Ed.), *Simple heuristics that make us smart* (pp. 209–234). New York: Oxford University Press.
[6] Brown, N. R. (2002). Real-world estimation: Estimation modes and seeding effects. In *Psychology of Learning and Motivation* (pp. 321–359, Vol. 41). Elsevier.
[7] Brown, N. R., & Siegler, R. S. (1993). Metrics and mappings: A framework for understanding real-world quantitative estimation. *Psychological Review*, *100*(3), 511–534.
[8] Nosofsky, R. M. (1984). Choice, similarity and the context theory of classification. *Experimental Psychology: Learning, Memory, and Cognition*, *10* (1), 104–114.

Link to Poster